Roll No. .....

TCE–404

B. TECH. (CE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

CONCRETE TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain with a neat sketch the phenomenon "Bulking of Sand". (CO1)

OR

(b) Enlist *four* requirements of good fine aggregate. (CO1)

2. (a) Enlist the various concrete operation in sequence and explain any *four* in detail. (CO2)

OR

(b) Explain different methods of curing of concrete. (CO2)

3. (a) List out major filed test carried out on cement. (CO1)

OR

(b) List out major test carried out on cement in laboratory. (CO1)

P. T. O.

4. (a) List out the provisions of IS 456 : 200 for water to be used in concrete. (CO3)

OR

(b) Classify the aggregates based o shape and texture. (CO3)

5. (a) "Water is the friend and enemy of concrete." Explain this statement with example. (CO2)

OR

(b) List out the various factors affecting workability of concrete. (CO2)

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TCE–406

B. TECH. (CE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ENGINEERING GEOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write a short note on "Mohs scale of hardness with examples". (CO1)

OR

(b) Enlist and explain the subdivisions of Geology. (CO1)

2. (a) Draw and explain any *five* different forms or structure of minerals. (CO1)

OR

(b) Describe the internal structure of the Earth with the help of a neat diagram. (CO1)

3. (a) Compare and contrast the following pairs : (CO1)

(i) Lava and Magma

(ii) Sills and Dykes

(iii) Concordant and Discordant bodies

P. T. O.

OR

(b) What are the various requirements of a good building stone ? (CO1)

4. (a) Distinguish between igneous, sedimentary, and metamorphic rocks. (CO2)

OR

(b) What is metamorphism ? Discuss the various agents of metamorphism along with some example. (CO2)

5. (a) Define the following terms : (CO2)

(i) Outcrop

(ii) Bedding or Stratification

(iii) Dip

OR

(b) Briefly explain the classification of Igneous rocks. (CO2)

Roll No. .....

TCE–401

B. TECH. (CIVIL ENGINEERING) (FOURTH SEMESTER) MID SEMESTER EXAMINATION, April, 2023

HYDRAULICS AND HYDRAULIC MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive the condition for the triangular channel of best section. (CO1)

OR

(b) A flow of water of 100 litres per second flows down in a rectangular channel of width 600 mm and having adjustable bottom slope. If Chezy's constant C is 56, find the bottom slope necessary for uniform flow with a depth of flow of 300 mm. Also find the conveyance K of the flume. (CO1)

2. (a) What do you mean by channel transition ? Explain the need of channel transition with examples. (CO1)

OR

(b) A trapezoidal channel having the side slope equal to 60° with the horizontal and laid on a slope of 1 in 750, carries a discharge of 10 m^3/s. Find the width at the base and depth of the flow for the most economical section. Take Chezy's C = 66. (CO1)

P. T. O.

3. (a) Draw a neat sketch of specific energy curve for an open channel with constant discharge and show the regions of subcritical, critical and supercritical flow on it. (CO1)

OR

(b) Water flows at the rate of 16 m^3/s in a channel 10 m wide at a velocity of 1.6 m/s. Calculate the specific energy head and critical depth. (CO2)

4. (a) State the classification of channels on the basis of bed slope. Draw the water surface profile for mild bed slope. (CO2)

OR

(b) A tidal bore is observed to move at a velocity of 6.0 m/s upstream in a wide river of depth 2.0 m. If the depth of flow after the passing of the bore is 4.25 m, estimate the velocities in the river before and after the bore. (CO2)

5. (a) Show that the brink depth is two-third of critical depth in an open rectangular channel. (CO2)

OR

(b) In a rectangular channel of 0.5 m width, a hydraulic jump occurs at a point where depth of flow is 0.15 m and Froude no. is 2.5. Determine the specific energy, critical depth, sequent depth and head loss. (CO2)

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TCE–402

B. TECH. (CE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

STRUCTURAL ANALYSIS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

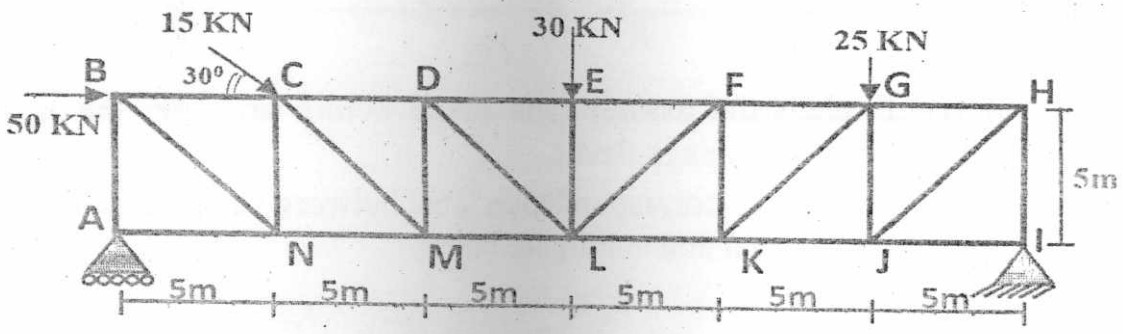
(ii) Each sub-question carries 10 marks.

1. (a) Present your understandings behind the assumptions made in statically determinate plane truss analysis. (CO1)

OR

(b) Establish the conditions for judging the given 2D and space trusses for statical determinacy. (CO1)

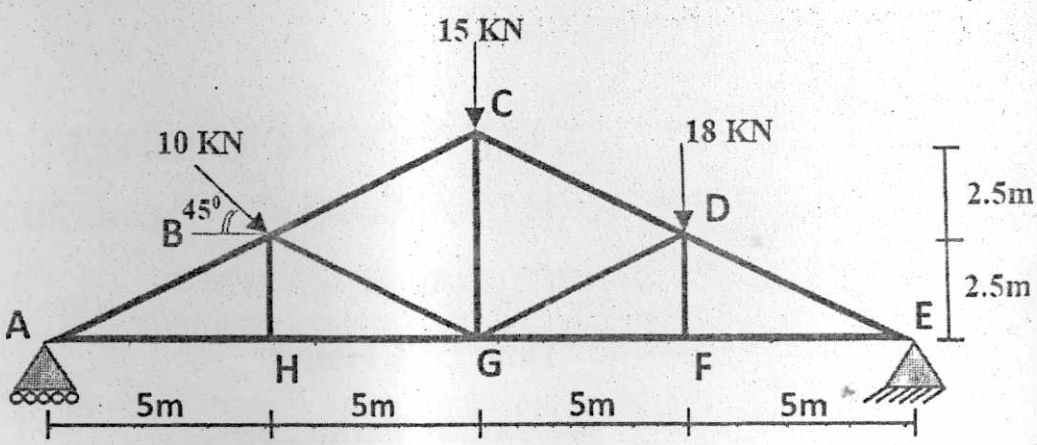
2. (a) Determine the member forces in GF, GK, JK, JI and HI given Flat truss by method of sections : (CO1)



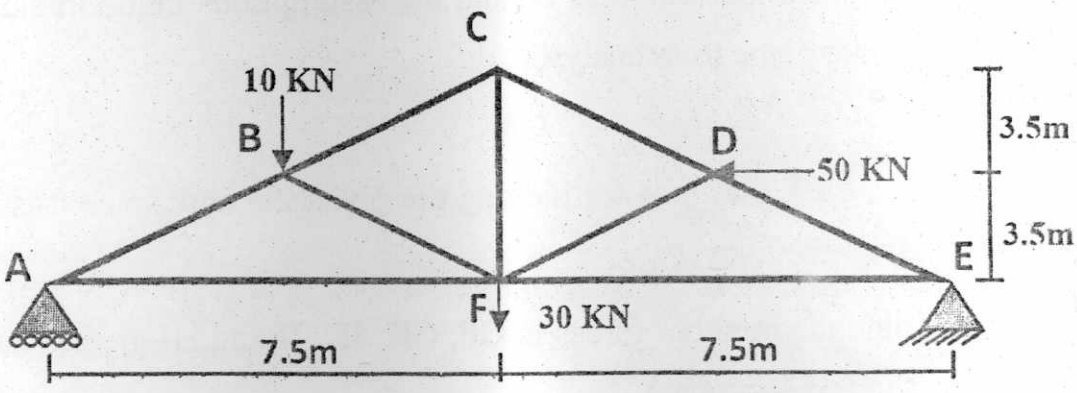
P. T. O.

OR

(b) Determine the member forces in the given Howe truss by method of tension coefficients : (CO1)



3. (a) Prove that the member forces of statically determinate Queen post truss are same by method of joints and tension coefficients method : (CO1)



OR

(b) (i) Explain the assumptions made before applying the methods of beam analysis in detail.

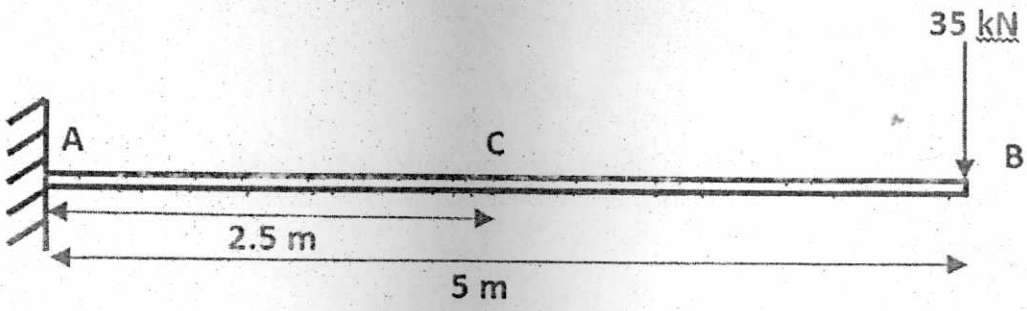
(ii) Present the correspondence table between supports, loadings, etc. of the original and conjugate beam. (CO2)

4. (a) Prove that slope at the free end of a cantilever beam of length L with concentrated moment M at free end is ML/EI, using Moment Area method ? (CO2)

OR

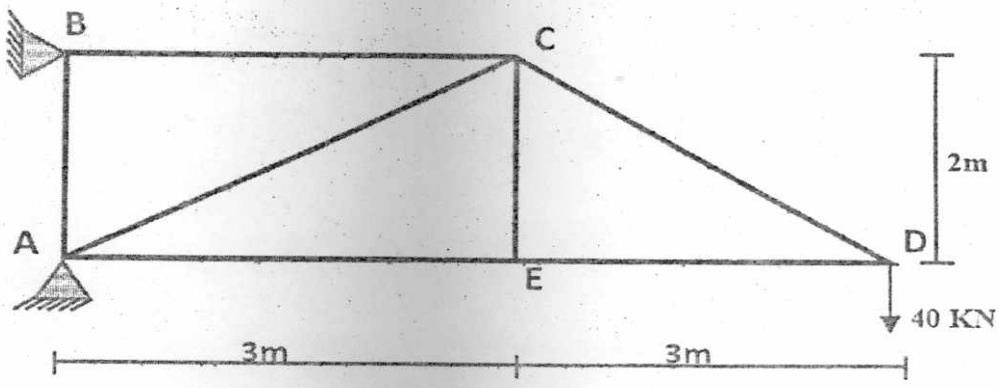
(b) Prove that deflection at the center of a simply supported beam of length L with concentrated load P at Centre is PL using Conjugate Beam method ? (CO2)

5. (a) Determine the deflection at the mid span for the given beam using Moment Area Method : (CO2)



OR

(b) Determine the vertical deflection at the loaded joint using Strain Energy Method : (CO2)



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TCE–405

B. TECH. (CIVIL ENGINEERING) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

GEOTECHNICAL ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following soil types : (CO1)

(i) LOESS

(ii) BENTONITE

(iii) PEAT

(iv) VARVED CLAY

OR

(b) In a shrinkage limit test, a container of volume 9.5 cc was filled with soil slurry. The weight of the soil slurry was 17.45 gm. The slurry was then gradually dried in air and then in oven at a constant temperature of 110°C. The Weight and volume of the dried soil were respectively 11.60 gm and 5.25 cc, respectively. Determine the shrinkage limit of the soil. (CO1)

P. T. O.

2. (a) Develop the following relationships : (CO1)

(i) $wG_s = Se$

(ii) Bulk unit weight in terms of water content, void ratio

(iii) Between void ratio and porosity

(iv) Dry unit weight, bulk unit weight and water content

OR

(b) 500 gm dry soil was subjected to sieve analysis and the following observations were made in terms of sieve size and weight retained :

4.75 mm — 10 g

2.00 mm — 165 g

1.00 mm — 100 g

0.425 mm — 85 g

0.212 mm — 40 g

0.150 mm — 30g

0.075 mm — 50 g

Calculate the coefficient of curvature and coefficient of uniformity values. (CO1)

3. (a) Explain the step by step procedure of determining the water content of the soil using pycnometer method. Develop expression for obtaining water content in terms of measurement made in pycnometer test. (CO1)

OR

(b) A 1000 cc core cutter weighing 950 gm was used to find out the in situ unit weight of embankment. The weight of the core cutter filled with soil was noted to be 2750 gm. Laboratory tests on the sample indicated water content of 12% and specific gravity of the solids as 2.65. Determine the bulk unit weight, dry unit weight, void ratio and degree of saturation of embankment soil. (CO1)

4. (a) Develop expression for estimating the coefficient of permeability of layered soil : (CO2)

(i) When flow is parallel to stratification.

(ii) When flow is normal to stratification.

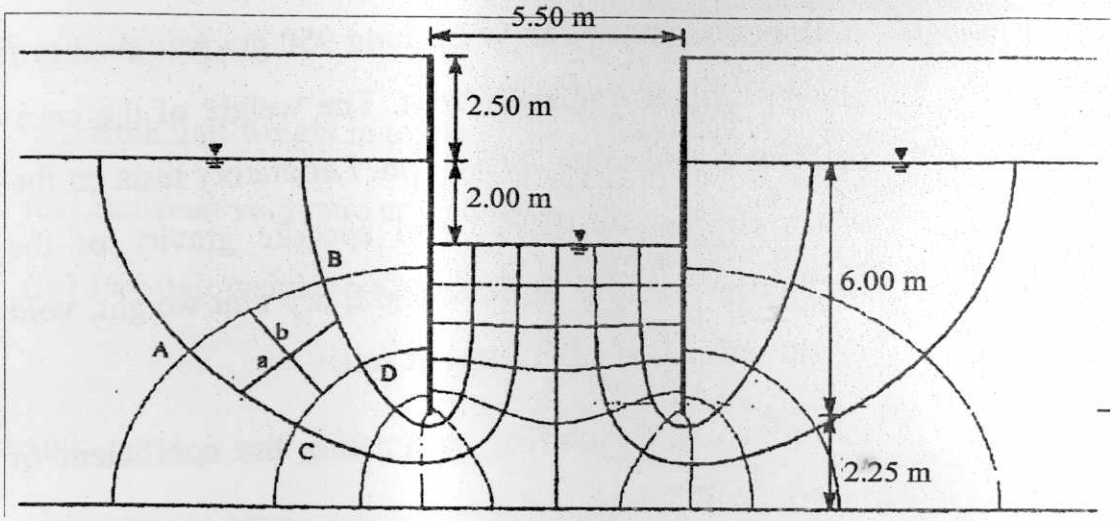
OR

(b) A pumping test was carried out at a site where a 10 m thick clay stratum overlies a fine sand stratum 2.5 m thick. Below the sand stratum, there lies an impermeable rock stratum. After a steady state condition was established, the pumping rate from a boring well was measured at 20,000 cc/sec. The average water levels in the two observation wells made at 6 m and 15 m radius were respectively, 5.0 m and 4.5 m below ground surface. Determine the coefficient of permeability of the sand. (CO2)

5. (a) Develop expression of Laplace's equation for two-dimensional seepage flow in the soil. State the assumptions involved in developing the equation. Mention methods used for solving Laplace equation. (CO2)

OR

(b)



Estimate the total seepage discharge around the sheet pile contributed from both side, if the coefficient of permeability is 5.50×10^{-8} m/sec. Aspect ratio is 0.85. (CO2)

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TCE–403

B. TECH. (CE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ADVANCED SURVEYING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain triangulation figures and layouts with the help of neat sketches. (CO1)

OR

(b) Explain the various types of signals used in triangulation survey. (CO1)

2. (a) From a satellite station S, 12.25 metres to the west of the main station B, the following angles were measured :

Angle BSC = 76°25′32″

Angle CSA = 54°32′20″

The stations S and C are to the opposite sides of the line AB. Calculate the correct angle ABC if the lengths AB and BC are 5286.5 and 4932.2 m respectively. (CO1)

P. T. O.

OR

(b) Define Triangulation. Describe the principle and objectives of triangulation. (CO1)

3. (a) Discuss the various points that should be considered while selecting the triangulation station. (CO1)

OR

(b) The following are the observed values of an angle : (CO2)

Angle	Weight
30°24′20″	2
30°24′18″	2
30°24′19″	3

Calculate :

(i) Probable error of single observation of unit weight

(ii) Probable error of weighted arithmetic mean

(iii) Probable error of single observation of weight 3

4. (a) Define the following terms : (CO2)

(i) Conditioned Quantity

(ii) Indirect Observation

(iii) True error

(iv) Most probable error

(v) Most probable value

OR

(b) Discuss various types of errors in surveying measurements. Give *one* example of each. (CO2)

5. (a) Calculate the standard error of the volume of a cuboid whose sides a, b and c have the values and standard error as under : (CO2)

$$a = 50 \pm 0.04 \text{ cm}$$

$$b = 40 \pm 0.03 \text{ cm}$$

$$c = 30 \pm 0.02 \text{ cm}$$

OR

(b) The following are mean values observed in the measurement of three angles α, β and γ at one station : (CO2)

$$\alpha = 76° \, 42' \, 46.2'' \text{ with weight } 4$$

$$\alpha + \beta = 134° \, 36' \, 32.6'' \text{ with weight } 3$$

$$\beta + \gamma = 185° \, 35' \, 24.8'' \text{ with weight } 2$$

$$\alpha + \beta + \gamma = 262° \, 18' \, 10.4'' \text{ with weight } 1$$

Calculate the most probable value of each angle.